

The Acquisition–Authority Envelope for Open-World Scientific Discovery

Working framework draft

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Abstract

Open-world scientific discovery is not a ranking problem alone. It is a joint control problem over route invention, evidence acquisition, question-conditioned processing, preservation of future search options, and scientific closure. We formalize this larger object as an *acquisition–authority envelope*: the set of discovery, utility, optionality, safety and cost vectors attainable by a class of controllers under a common resource and source-access contract. The formulation absorbs strong lexical, reasoning-aware, structure-aware, citation-expansion, coverage, value-of-information and verification-aware stopping mechanisms as interchangeable donor components rather than defining the contribution against any one retriever. Four analytic results locate the remaining problem. First, an acquisition-ceiling theorem proves that no downstream ranker, reader or synthesizer can recover decision-relevant evidence that no admissible route emitted. Second, a necessary-and-sufficient factorization theorem shows that a donor product can implement correct scientific closure exactly when closure truth is constant on every fibre of its observable state. Third, an indistinguishable-world lower bound shows that a controller cannot simultaneously guarantee low false closure and high valid closure yield while a material alternative world remains observationally unresolved; acquisition must refine the state or the system must remain open. Fourth, the resulting obligation completion is maximally permissive among universally sound closure rules under the declared contract.

The archived experiments test parts of this theory at different levels. In a frozen 390-task complete-gold controlled index, the governed system reaches mean recall 0.979487 versus 0.666667 for the strongest protocol-frozen confirmatory comparator and makes no premature task closures, but the frozen plan requires the result to remain underpowered and descriptive. In a separate external adverse campaign, the governed routes emitted only 7 of 229 gold identifiers, making candidate generation—not downstream selection—the binding ceiling; that terminal remains adverse. On 400 protected heterogeneous exact contracts, typed authority obtains 400/400 exact closure decisions versus 250/400 for a donor-complete product that omits unresolved material routes, while an information-equivalent ideal product ties 400/400 as the factorization theorem predicts. These results establish an analytic and mechanistic framework, not deployed open-world superiority. A separately frozen 768-cluster, four-arena successor is the prospective test of cross-arena acquisition-envelope superiority; its external worlds, comparator reproductions and independent custody remain unbound, and no outcome is claimed.

Keywords: open-world scientific discovery; information retrieval; deep research agents; acquisition envelope; scientific closure; systematic review; stopping rules.

1 Introduction

Scientific discovery agents increasingly combine retrieval, query reformulation, reranking, graph expansion, selective reading, state compression, contradiction handling and learned stopping. Treating any one component as “the discovery system” makes scientific comparison unstable: a new

retriever can dominate the old acquisition layer without answering whether the assembled controller found the decision-relevant evidence, kept useful future routes alive, or closed the scientific task safely. We therefore ask a larger question: *what joint discovery frontier is attainable under a fixed source-access and resource contract, and what information is necessary for a point on that frontier to carry scientific-closure authority?*

The paper makes five contributions. First, it defines the *acquisition–authority envelope*, a non-compensatory frontier over decision-relevant discovery, scientific utility, future-search optionality, false closure, valid closure yield and total charged cost. This moves the unit of theory above individual retrievers and stopping rules. Second, it proves an acquisition-ceiling theorem that attributes an adverse outcome to route generation whenever the required evidence never entered the candidate pool. Third, it gives a necessary-and-sufficient state-factorization criterion for scientific closure and an indistinguishable-world lower bound for any controller whose observations conflate closable and materially open worlds. Fourth, it constructs the maximally permissive fail-closed completion of a donor controller relative to a declared obligation contract. Fifth, it instantiates the theory with earned route independence and question-conditioned read state, then places all archived positive, adverse and undetermined experiments at their exact authority levels rather than averaging them into one apparent win.

Let route r be characterized by a backend identity b_r , query-derivation process q_r , and capture/content identity policy c_r . A distinct route label is insufficient for independence. Evidence for route independence must be derived from the actual generation and capture process rather than asserted as a boolean.

2 Question-conditioned cumulative read state

A paper being previously encountered is weaker than the statement that it has been processed for the current research question, extraction schema and content version. Content identity is therefore held separately from question-conditioned read state. A changed question or materially changed content can legitimately reopen a source for processing without pretending that it is newly discovered evidence.

3 Route stopping versus task stopping

Route outcomes such as continue, reformulate, switch, budget-stop, unavailable and locally flat are separated from task-level scientific closure. A route may stop because its marginal yield is low while another route is still untried, unavailable or censored. Coverage diagnostics therefore stay advisory; they cannot promote global completeness.

This fail-closed rule is especially important for capture–recapture-style diagnostics. Adaptive search routes are not automatically independent capture occasions, and zero overlap cannot justify a confident bounded estimate of unseen literature.

4 Acquisition is not closure

The first design move is to separate two problems that are often collapsed into one policy. An *acquisition policy* decides what to search, retrieve, inspect, expand or reread next. A *closure authority* decides whether the evidence state is allowed to support route-local stopping or scientific task completion. The larger theory then reconnects these layers through a joint frontier: better acquisition may improve discovery, utility and optionality, but it improves scientifically sound

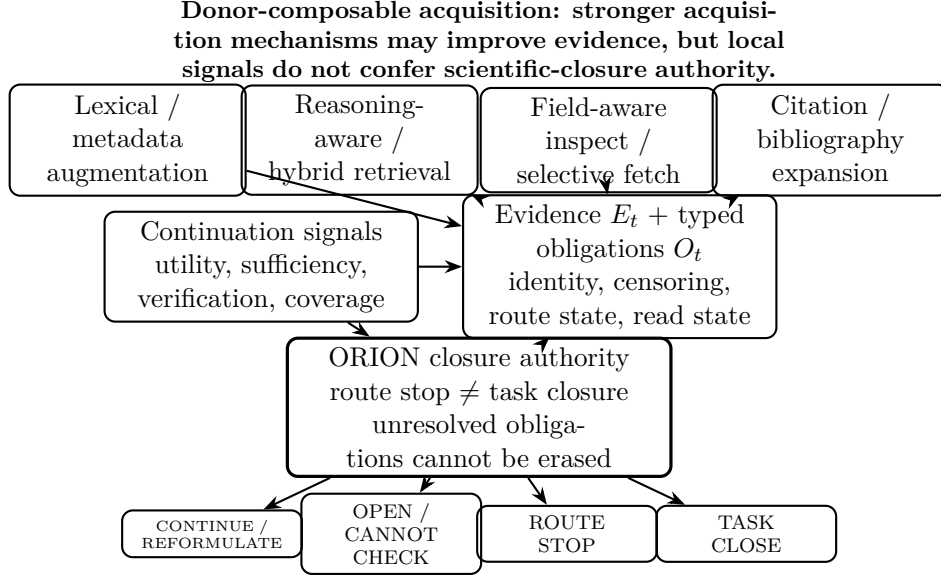


Figure 1: **Figure P2-7.** Acquisition mechanisms are intentionally composable: lexical/metadata augmentation, reasoning-aware or hybrid retrieval, field-aware selective fetching, citation expansion and continuation signals may all improve what is acquired. Their outputs update evidence and typed obligations, but a separate authority layer controls route stop, open and undetermined states and task closure. The figure therefore locates the residual at the meaning of acquisition outcomes rather than at a claim to have invented the strongest retriever.

closure only when it discharges an obligation or exposes a state distinction that closure needs. This makes strong external retrieval and stopping mechanisms composable rather than adversarial to the contribution.

Let x_t denote the observable search state at step t . We write an acquisition mechanism as

$$a_t \sim A(x_t),$$

where a_t may be a lexical query, a newly generated source route, a reasoning-aware query, a fielded Boolean filter, a graph-expansion action, a selective fetch, a reranking operation, a state-preservation/recovery action, or a request for another retrieval round. No novelty is claimed here for generic members of that action family. Instead, acquired evidence updates both an evidence state E_t and a typed obligation state O_t . Closure is a distinct map

$$C(E_t, O_t) \in \{\text{CONTINUE, ROUTE-STOP, OPEN, CANNOT-CHECK, TASK-CLOSE}\}.$$

A utility, relevance, confidence, sufficiency or coverage signal may influence A and may recommend a local stop, but it cannot by itself delete an unresolved obligation in O_t . This interface is where the residual contribution of this paper sits.

4.1 Absorbing strong acquisition mechanisms

The acquisition layer is intentionally donor-composable. SAGE shows that keyword-oriented agent queries can make strong lexical retrieval difficult to beat and that corpus-side metadata/keyword augmentation can further improve retrieval [6]. AgentIR shows that an agent’s reasoning trace can be used jointly with its query as a retrieval signal [3]. SIEVE shows that fielded Boolean admission,

structure-rich inspection and selective section fetching can improve the accuracy–context tradeoff [13]. Breadth-first bibliography expansion can recover literature missed by API-only retrieval, while also demonstrating why citation-list recall cannot be the only evaluation axis [12]. Lexical, metadata-augmented, reasoning-aware, field-aware and citation-expansion retrieval are therefore treated as candidate strengths to compose, not as inventions of this work.

The same rule applies to continuation and stopping signals. DeepControl uses marginal information utility to regulate continued retrieval [16]; decision-theoretic policies optimize screening stops against explicit costs and payoffs [5]; HALT uses claim–evidence coverage to verify whether expected evidence has been obtained [11]; MiCP applies conformal prediction to adaptive multi-turn reasoning to maintain a target coverage guarantee while permitting early stopping [18]; and structured Search-R1 stopping judgments explicitly encode sufficiency and remaining gaps [8]. These signals and guarantees can all be useful inputs to acquisition or route-local stopping. The question asked here is higher-order rather than a competition over any one signal: what joint discovery frontier does their composition attain, and is the resulting state sufficient to certify closure when routes are dependent, unavailable, censored or only locally exhausted?

4.2 The authority residual

This decomposition yields four interface requirements and one wider comparison claim. Route independence must be earned from backend, query-derivation and capture identity rather than from tool names; route-local stopping and task-global closure occupy different authority levels; unavailable or censored routes remain typed obligations rather than empty results; and content identity stays distinct from question-conditioned processing state. Above those requirements, the scientific object is the acquisition–authority envelope: the attainable joint frontier over discovery, utility, optionality, closure safety, valid closure yield and charged cost. Section 6 derives when a donor-complete product can reach and authorize a point on that frontier.

The resulting design is deliberately asymmetric. It is permissive about *how* evidence is acquired: a stronger retriever, query generator, expansion rule or verification signal can replace a weaker one. It is conservative about *what acquisition outcomes are allowed to mean*: local flatness, high estimated coverage, high confidence, apparent sufficiency or a provider failure cannot silently become proof of global completeness. This asymmetry is what permits the paper to absorb progress in retrieval without making its claimed authority boundary obsolete whenever a stronger retriever is published.

4.3 Empirical consequence

The decomposition also fixes the widening experiment. Candidate-generation methods must be compared and composed under matched *admissible exposure*, not merely equal request counts. Only after the acquisition ceiling is high enough to make processing and stopping observable should the state and closure layers be challenged with direct ablations and strong donor products. A positive external result must therefore improve the guarded joint frontier; a cheaper stop that hides reachable evidence or an attractive report assembled from a truncated candidate pool is not a positive result. Conversely, a stronger donor retriever that improves acquisition while leaving the closure constraints intact strengthens the composed system and raises the comparison standard rather than weakening the theory.

5 Formal mechanics

This section fixes the objects the evaluation measures. Every definition below is implemented in the system under test rather than assumed: work identity and read state, route independence and coverage, and route control each correspond to a module of the implementation, named in Section 13. Where a definition is deliberately conservative we state the failure it is conservative against, because the direction of the bias is what makes the refusal a safety property rather than a preference.

5.1 Work identity and content identity

Let R be the set of raw identifiers a route can emit (DOIs, arXiv ids, OpenAlex ids, local ids, URLs). Canonicalization is a map $\varphi : R \rightarrow \mathcal{S} \times V$ sending a raw identifier to a scheme-qualified key and a rendition version, so that $\varphi(\text{arXiv:2301.00001v1})$ and $\varphi(\text{arXiv:2301.00001v2})$ share a key and differ only in V : a version is a rendition of one work, not a second work. A declared scheme is treated as a claim rather than a credential; an identifier that does not match the shape of the scheme it announces resolves to UNKNOWN instead of entering the graph as a key that can never be fetched.

A *work* is the alias-closure of canonical keys. Given identity records with alias sets $A_1, \dots, A_n$, works are the connected components of the graph on $\bigcup_i A_i$ that joins two keys sharing a record. Closure is computed as a projection over an append-only ledger rather than at write time, because an alias learned late can join two works recorded early; folding at write time would freeze the wrong partition.

Separately from work identity, each retrieved rendition carries a content digest c . Route overlap, deduplication and coverage are computed over content digests, never over retrieval identifiers: a source that re-mints identifiers would otherwise make two routes appear disjoint when they retrieved the same paper, which inflates every coverage quantity built on their overlap.

5.2 Question-conditioned read state

A read receipt is a tuple (w, c, σ, f) : work, content digest, extraction schema version, and frame f — the scientific question the extraction was performed for. Given the receipt set $\mathcal{R}$, the read decision for a candidate (w, c, σ, f) is

$$\text{read}(w, c, \sigma, f) = \begin{cases} \text{UNSEEN} & \nexists r \in \mathcal{R} : w_r = w \\ \text{CONTENT_CHANGED} & \exists r : w_r = w, \nexists r : (w_r, c_r) = (w, c) \\ \text{NEW_SCHEMA} & \exists r : (w_r, c_r) = (w, c), \nexists r : (w_r, c_r, \sigma_r) = (w, c, \sigma) \\ \text{NEW_FRAME} & \exists r : (w_r, c_r, \sigma_r) = (w, c, \sigma), \nexists r : r = (w, c, \sigma, f) \\ \text{ALREADY_READ} & \exists r \in \mathcal{R} : r = (w, c, \sigma, f). \end{cases}$$

ALREADY_READ therefore requires agreement on all four coordinates. This yields the distinction the evaluation depends on: a repeat extraction with all four coordinates equal is *duplicate processing* and is waste, while a repeat extraction under a changed content digest, schema or frame is a *legitimate reread* and is required work. Collapsing the two — treating “this paper has been seen” as “this paper has been read for this question” — is how a system reports coverage of material it only skimmed for a different purpose.

5.3 Earned route independence

A route capture is a tuple $\rho = (k, b, d, C)$: route kind, backend identity, query-derivation process, and the set C of captured content digests. For two captures ρ_i, ρ_j the independence predicate is

$$\text{ind}(\rho_i, \rho_j) \iff k_i \neq k_j \wedge b_i \neq b_j \wedge d_i \neq d_j,$$

and the verdict otherwise names the shared cause (SAME_ROUTE_KIND, SHARED_BACKEND, SHARED_QUERY_DERIVATION). Independence is thus *earned* from the generation and capture process; a distinct route label carries no evidential weight.

Proposition 1 (Direction of the conservatism). *Positive dependence between capture occasions biases unseen-mass estimators downward. Admitting a dependent pair therefore produces false confidence that search is near exhaustion, whereas refusing an independent pair only forfeits a diagnostic. The predicate is accordingly biased toward refusal.*

5.4 Coverage diagnostics are never authority

For an independent pair with captures of size n_i, n_j and overlap m_{ij} , the two-source estimate is $\hat{N} = n_i n_j / m_{ij}$, undefined at $m_{ij} = 0$. Zero overlap places no upper bound on the population, so the system returns a refusal with reason rather than a large number that would read as a measurement. When several independent pairs yield estimates, the reported value is $\max_{ij} \hat{N}_{ij}$: the largest population implies the most still unfound, which is the direction that fails safe. Every coverage estimate is marked diagnostic-only, and adaptive search is explicitly not a sequence of independent capture occasions — following up on what was just found converts singletons into doubletons and deflates the estimator toward “nearly done”.

5.5 Route control state machine

Let a route hold attempts $a_1, \dots, a_t$, each with retrieved digests D_i , novel digests $N_i \subseteq D_i$, cost, failure signatures and a backend-exhaustion flag. Marginal novelty of an attempt is $|N_t|/|D_t|$, and the flat streak z is the number of trailing attempts that succeeded, reported no exhaustion and returned no novel digest. Under a frozen policy with thresholds $\theta_{\text{ref}} \leq \theta_{\text{sw}}$, the decision is taken in strict priority order:

$$\delta = \begin{cases} \text{STOP (budget)} & \text{remaining budget} \leq 0 \\ \text{STOP (exhausted)} & a_t \text{ reports backend exhaustion} \\ \text{SUSPEND} & a_t \text{ carries execution-failure signatures} \\ \text{SWITCH} & z \geq \theta_{\text{sw}} \wedge \text{alternatives exist} \\ \text{REFORMULATE} & z \geq \theta_{\text{ref}} \\ \text{CONTINUE} & \text{otherwise.} \end{cases}$$

Two separations in this table are load-bearing. First, backend exhaustion and transport failure are distinct states: the former is a statement about the index, the latter about the channel, and a system that merges them converts an outage into evidence of absence. The latter yields SUSPEND, a non-terminal state carrying an open obligation, not STOP. Second, each terminal decision carries typed reopen triggers (budget increase, backend recovery or refresh, material frame change, material finding on an alternative route); a route reopens exactly when an observed trigger intersects that set.

Proposition 2 (Fail-closed stopping). *No route decision certifies task saturation. In the implementation every route decision, STOP included, carries an explicit declaration that it does not certify task saturation, so route-local exhaustion cannot be lifted to task-level completeness by any composition of route outcomes.*

Task closure is therefore a separate judgement over the route ensemble. It requires that every route is in a terminal or explicitly reopenable state, that no suspended or unavailable route leaves an unmet obligation, and — by Proposition 2 — that no coverage diagnostic has been used as its authority. An unavailable route remains an open coverage obligation and is reported as such; it is never converted into evidence that nothing further exists.

5.6 What the evaluation must therefore measure

The definitions above fix four measurable error families, each with a ground-truth condition that only a complete-gold world supplies: route-stop false positives (a route stopped while it could still reach unfound relevant items), route-stop false negatives (a route continued past its exhaustion point), premature task closure (task closed while a gold item was still reachable within remaining budget), and duplicate processing versus legitimate reread as separated by the four read coordinates. Route overlap, per-route marginal contribution and earned-independence status are computed over content digests. Metrics whose denominator is the gold set are reported only where that denominator is complete; the conditions are stated per metric in the frozen measurement plan.

6 The acquisition–authority envelope

The acquisition/closure split becomes substantially more useful when it is lifted from a design rule to a comparison theory. A scientific discovery controller has four coupled responsibilities: generate or select a route, acquire evidence through it, convert acquired material into decision-usable state while preserving useful future cues, and decide whether the task may close. A retriever improves only one part of this object; a stopping policy can save cost while making another part worse. This section defines the joint object and derives the conditions under which such local improvements can be composed into a scientifically stronger controller.

6.1 Worlds, obligations and a non-compensatory frontier

Let $w \in \mathcal{W}$ be a scientific world and let h_t be the complete observable transcript after charged step t . The task contract assigns to w a set $\mathcal{O}(w)$ of *material scientific obligations*. An obligation may be a claim that needs evidence, a contradiction that needs both sides inspected, a dataset or code artifact needed to establish reproducibility, a route whose availability must be resolved, or a previously retrieved source that must be processed under the current question. Materiality means that resolving the obligation can change the predeclared decision, conclusion, uncertainty state or admissible closure state. It is not synonymous with relevance to a search query.

A controller π comprises route generation G , acquisition A , a question-conditioned state compiler M , and closure authority C . Under budget B , source-access contract $\mathcal{I}$ and task distribution P , let its outcome vector be

$$Z_B(\pi, w) = (D, U, V, -F, Y, -K).$$

Here D is decision-relevant evidence discovery, U is final scientific utility, V is the value of search options preserved for prospectively hidden future questions, F is false task closure, Y is valid

closure yield, and K is fully charged cost. Signs are chosen so larger is better except that F is a non-compensatory safety coordinate: a gain in report quality cannot purchase permission to hide a false closure.

Definition 1 (Acquisition–authority envelope). For an admissible policy class Π ,

$$\mathcal{E}_{B,\mathcal{I}}(\Pi) = \text{cl} \{ \mathbb{E}_{w \sim P}[Z_B(\pi, w)] : \pi \in \Pi, \mathbb{E}[K(\pi, w)] \leq B \}.$$

Its *guarded frontier* retains only points satisfying the prospectively fixed false-closure, valid-closure-yield, provider-validity and cost guards. A candidate establishes envelope improvement only by lying outside the guarded frontier of the frozen donor-complete class under the same P , B and $\mathcal{I}$; winning against a conveniently weak member of that class is insufficient.

This definition absorbs the strongest neighbouring components instead of opposing them. Lexical and corpus-side augmentation, reasoning-aware retrieval, fielded inspection and selective fetching, and citation expansion enlarge the action family of G and A [6, 3, 13, 12]. Claim–evidence coverage, marginal information utility, conformal continuation, decision-aware and evidence-aware stopping enlarge the policy and signal class [11, 16, 18, 5, 7, 1]. Finite source certificates, obligation graphs and question-conditioned memory enlarge M and the observable closure state [2, 17, 9, 4]. These are donor territory. The residual question is whether their composition reaches a better guarded frontier and whether its closure decision is identifiable from the state the product actually retains.

Proposition 3 (Donor-monotonicity). *If $\Pi_1 \subseteq \Pi_2$ under identical task, access and budget contracts, then $\mathcal{E}_{B,\mathcal{I}}(\Pi_1) \subseteq \mathcal{E}_{B,\mathcal{I}}(\Pi_2)$. Consequently, adding a sound donor action, signal or memory operation cannot make the attainable envelope smaller, though it may leave the guarded frontier unchanged after its cost is charged.*

Proof. Every policy feasible in Π_1 remains feasible in Π_2 ; take the closure of their attainable outcome vectors. The guarded statement follows because the same non-compensatory filters are applied to the superset. $\square$

This elementary monotonicity result changes the novelty test. A new paper that improves retrieval should be absorbed into the comparator class. The theory is falsified, not protected, if the expanded donor class reaches every proposed candidate point under matched resources.

6.2 The acquisition ceiling: why seven emitted identifiers matter

Let G_w be a complete evaluator gold set for a fixed task world, $A_\pi(w)$ the union of gold-normalized identities ever emitted by any admissible route during a frozen run, and $S_\pi(w)$ the final submitted set. A downstream ranker, reader or synthesizer that is forbidden to invent evidence satisfies $S_\pi(w) \cap G_w \subseteq A_\pi(w) \cap G_w$.

Theorem 1 (Run-conditional acquisition ceiling). *For any such downstream policy and any nonempty G_w ,*

$$\text{Recall}(S_\pi; G_w) \leq \frac{|A_\pi(w) \cap G_w|}{|G_w|}.$$

More generally, if obligations have nonnegative weights μ_o and each obligation can be discharged only by at least one member of evidence set E_o , then the unnormalized resolved-obligation weight is at most the total weight of obligations for which $A_\pi(w) \cap E_o \neq \emptyset$.

Proof. The submitted relevant set is a subset of the acquired relevant set, so its cardinality cannot be larger. For the weighted form, an obligation whose required evidence set is disjoint from the acquired pool cannot be discharged by a downstream operation over that pool; summing the remaining nonnegative weights gives the bound. $\square$

The theorem separates candidate-generation failures from selection failures without assuming that a particular retriever is optimal. In the archived external adverse campaign, all routes emitted only 7 of the 229 gold identifiers; six were submitted and one was discarded. The run-conditional recall ceiling was therefore $7/229 = 0.030568$, and 222 gold identities were unavailable to every downstream selection rule on that capture. This does not estimate general web discoverability. It proves the narrower but decisive causal fact that reranking or synthesis could not repair that run. The adverse terminal and the separate provider-validity failure remain unchanged.

The same analysis exposes why matched request counts are not a matched scientific opportunity. Let $\xi(a, G_w)$ be the admissible exposure of action a : the prospectively validated opportunity, through the scorer-compatible interface, to emit identities in G_w . Equal $\sum_a 1$ does not imply equal $\sum_a \xi(a, G_w)$.

Proposition 4 (Call matching does not identify exposure matching). *For every positive call budget m , there exist two m -call policies under the same nominal budget whose run-conditional acquisition ceilings are respectively one and zero. Hence call-count equality alone gives no nontrivial bound on the difference in discoverability.*

Proof. Take a one-item gold set. Give the first policy m calls whose admissible support contains the item with a deterministic first-call return, and give the second m calls to an interface whose output support is disjoint from the gold identity namespace. Both spend m calls; their acquired-gold fractions are one and zero. $\square$

The archived campaign gave the lexical arm three calls to the only scoring-admissible backend and the governed arm one such call plus two inadmissible calls. Its three-versus-three request count was therefore not an exposure match. In a future protected study, exposure must be certified outcome-blind through interface-class witnesses or treated as CANNOT-CHECK; gold itself cannot be used as a worker-visible route oracle.

6.3 When can a product implement scientific closure?

Let $s : \mathcal{W} \rightarrow \mathcal{S}$ be everything a donor product makes available to its closure rule at a fixed decision time: acquired evidence, coverage and utility signals, route states, compiled memory and charged resource state. Write

$$\mathcal{Y}_C = \{\text{CLOSE}, \text{OPEN}, \text{CANNOT-CHECK}\}$$

and let $\kappa : \mathcal{W} \rightarrow \mathcal{Y}_C$ be the correct task decision under the declared scientific contract.

Theorem 2 (Closure factorization). *There exists a decision map $g : \mathcal{S} \rightarrow \mathcal{Y}_C$ such that $\kappa = g \circ s$ if and only if κ is constant on every fibre $s^{-1}(z)$.*

Proof. If $\kappa = g \circ s$ and $s(w) = s(w')$, then $\kappa(w) = g(s(w)) = g(s(w')) = \kappa(w')$. Conversely, if κ is constant on every fibre, define $g(z)$ to be $\kappa(w)$ for any w satisfying $s(w) = z$. Fibre constancy makes the definition independent of the chosen w . $\square$

The theorem gives an exact necessary-and-sufficient condition, not a claim that a particular architecture is intrinsically more expressive. Coverage, confidence and route-local stop signals are

sufficient for task closure only when no two worlds with the same retained state require different task decisions. A state that drops the materiality or unresolved status of an unavailable route creates a mixed fibre and cannot be repaired by a different classifier over that same state. An information-equivalent product that retains the missing coordinate has pure fibres and can tie. This is exactly the boundary exhibited by the protected exact-contract battery: the available-route product fails on unresolved material routes, while the ideal typed product is decision-identical to the authority rule.

6.4 The price of an observationally unresolved world

The factorization obstruction has a quantitative version. Consider two task worlds w_0, w_1 whose correct decisions are respectively CLOSE and OPEN. Let P_0 and P_1 be the distributions of the complete observable transcript available to the controller before its decision. Count CANNOT-CHECK as not yielding a valid closure in w_0 but not as false closure in w_1 .

Theorem 3 (Indistinguishable-world lower bound). *For every randomized closure rule, its false-closure probability F_1 in w_1 and valid-closure shortfall L_0 in w_0 satisfy*

$$F_1 + L_0 \geq 1 - \text{TV}(P_0, P_1).$$

In particular, if the two worlds are observationally identical, then $F_1 + L_0 = 1$ and $\max(F_1, L_0) \geq 1/2$.

Proof. The decision rule is a binary test between P_0 and P_1 once OPEN and CANNOT-CHECK are combined as “do not close.” The sum of its two testing errors is bounded below by one minus total variation. When $P_0 = P_1$, a transcript-dependent close probability has the same expectation in both worlds, so false closure plus failure to close the closable world equals one. $\square$

This lower bound prevents a cosmetic conversion of every negative result into a positive terminal. If the decisive route is absent from the observable action support, no stopping heuristic can create the missing distinction. Scientific progress must instead do at least one of three things: invent an acquisition action that makes P_0 and P_1 distinguishable, retain a state coordinate that an earlier compiler erased, or introduce and validate a substantive world assumption that rules out one member of the pair. Otherwise a calibrated open or CANNOT-CHECK state is the only honest output.

6.5 Authority completion and its maximality

For a transcript h , let $R(h)$ be the registered set of material obligations that are not yet independently discharged. Assume an *adversarial realizability contract*: for each $o \in R(h)$ there exists a task world consistent with h in which the evidence behind o changes the scientific decision or admissible closure state. Define the obligation-completed closure rule

$$C^+(h) = \begin{cases} \text{CLOSE}, & R(h) = \emptyset \text{ and the task contract is valid,} \\ \text{CANNOT-CHECK}, & \text{required observability or provider validity is unestablished,} \\ \text{OPEN}, & \text{otherwise.} \end{cases}$$

Theorem 4 (Maximally permissive sound completion). *Relative to the declared obligation and realizability contract, C^+ has no false closure. Moreover, any rule using the same contract that closes at a history where $R(h) \neq \emptyset$ is unsound on at least one compatible world. Thus no universally sound rule closes a strict superset of the histories closed by C^+ .*

Proof. If C^+ closes, every registered material obligation is discharged and the contract is valid, which is the contract’s definition of admissible closure. If another rule closes with $o \in R(h)$, adversarial realizability supplies a world consistent with the same history in which o changes the decision; closure is false in that world. Therefore every universally sound rule must refuse that history. $\square$

The theorem is broad but explicitly contract-relative. It does not claim that an open-web system can enumerate every material obligation from finite search. Completeness of $R(h)$ requires either a closed registry, a validated sampling model, independently curated hidden-route custody, or a declared CANNOT-CHECK boundary. The theorem nevertheless provides a positive composition result: any future acquisition, ranking, stopping or memory donor can be installed below C^+ without losing sound closure, and any improvement that discharges more obligations moves the controller toward valid closure rather than merely increasing a surrogate score.

Corollary 1 (When acquisition enlarges closure authority). *Adding a donor mechanism can expand discovery or reduce cost by Proposition 3. It enlarges the set of histories that can be soundly closed only if the resulting observation/state map splits a previously mixed closure fibre or discharges an obligation in $R(h)$.*

This is the sense in which the proposed theory envelops its nearest work. Retrieval, coverage, information utility, stopping, obligation certificates and memory are all admissible mechanisms inside the envelope. Their scientific value is measured by the joint frontier they make reachable; their outputs do not acquire global authority merely by being numerically confident.

6.6 A failure-derived candidate: obligation-directed adaptive exploration

The theory also specifies a materially stronger acquisition candidate rather than treating fail-closed authority as a substitute for finding evidence. Let $m_t(o)$ be the posterior unresolved mass of material obligation o , let $Q_t(a)$ be the prospectively evaluated option value that action a preserves for hidden future questions, and let $J_t(a)$ be the expected reduction in closure-fibre ambiguity—the probability mass of closable/open worlds that the action is expected to separate. An *obligation-directed adaptive exploration* (ODAE) policy chooses from route invention, retrieval, processing, contradiction resolution, state preservation/recovery and closure verification by a charged index of the form

$$a_t \in \arg \max_{a \in \mathcal{A}(h_t)} \frac{\text{LCB}_t \left[\Delta \sum_{o \in R(h_t)} \mu_o m_t(o) \mid a \right] + \lambda Q_t(a) + \beta J_t(a)}{\text{cost}_t(a)}.$$

The lower-confidence term prevents an uncertain route estimate from being treated as established gain; Q_t prevents current-question compression from destroying future route cues; and J_t values actions that make the closure decision identifiable even when their immediate document count is low. If no action has validated support through the frozen interface, the controller emits CANNOT-CHECK rather than assigning that route zero value.

Generic value-of-information, active search, coverage estimation and adaptive stopping are donor-owned. The proposed residual is their non-compensatory coupling to material scientific obligations, option preservation and closure fibre separation under one charged budget. ODAE is a prospective policy family, not an executed result or current general capability. Its direct falsifier is a donor-complete controller that reaches the same or a better guarded frontier on fresh tasks. Its causal tests must separately vary (i) route support, (ii) retrieved-but-unprocessed evidence, (iii) contradiction sides, (iv) future-query optionality and (v) closable clean cases; improvement on only the family that inspired the policy is insufficient.

6.7 A prospective cross-arena discriminator

The theory converts the historical adverse terminals into explicit research problems rather than deleting them. The 7/229 exposure result demands route invention and admissible-support estimation before reranking. The one-third scoring-eligible exposure mismatch demands source-interface matching rather than call-count matching. The retrieved-but-unprocessed and future-query failures demand option-valued state preservation. Provider-invalid and unresolved-route failures demand independent observability custody and an authority-complete state. Clean closable tasks remain a required counterweight against an always-open policy.

A separately frozen successor gives these problems a common empirical target. It specifies 768 source-disjoint task-world clusters—192 in each of four strata—across four arenas, identical admissible route exposure, three frozen primary comparators, an intersection requirement over every contrast, and a joint power target of 0.90 under its declared planning model. That design is a protocol, not a result. Archived baseline-task-world reproduction, four source-disjoint arena snapshots, a capability-matched external comparator, independent evaluator/gold custody, and live response/cost receipts are still required. Until those bindings exist and the protected campaign runs, the general cross-arena envelope-superiority hypothesis remains untested and the historical manuscript terminal remains unchanged.

7 Methods

The design choices in this section were fixed before the result-bearing offline campaign. Post-outcome edits to this prose are limited to correcting execution status (for example, saying that an external benchmark has not yet been run) and do not change a hypothesis, task family, system, metric, budget, exclusion rule, margin or decision rule. Machine-readable artifacts govern rather than this narrative: a design freeze, a measurement plan, a statistical plan, an external-artifact access audit, the controlled-world freeze and a prospective execution binding. Each is archived, and Section 13 names where.

7.1 Three-valued outcomes

Every decision this paper measures is three-valued rather than binary. A task may be determined complete, determined incomplete, or left *undetermined*; a route may be determined exhausted, determined to have more to give, or left undetermined. “We could not determine this” is recorded, reported and counted separately from “we determined this is false”, and the two are never merged. The distinction is the point: a provider outage, a missing credential or a censored route produces an undetermined outcome, and an undetermined outcome is not evidence of absence. Throughout the paper an outcome described as *undetermined* is this third value, carried in the evidence record and in every downstream count, never silently resolved to a negative.

7.2 Prospective discipline

The protocol is frozen and versioned before execution. A result-bearing execution is admissible only after subject code, data, evaluator, system identities, budgets and repeat seeds are bound. For the completed offline companion, the run manifest records that no outcome was accessed before the freeze, together with the exact subject revision, content hashes, 14 system identifiers, three repeat seeds and host-owned resource limits. The bound subject revision passed the full test suite before the first aggregate outcome snapshot was recorded. Publication summaries and manuscript prose

may be generated after outcomes, but they may only project the frozen records; they cannot change the scientific decision rules.

7.3 Evaluation environments

Three environments are distinguished because no single one supplies both a complete denominator and open-world realism. Only the first has a completed result-bearing campaign in this revision.

Offline complete-gold controlled index. Recall is only measurable where the relevant set can be enumerated. The controlled index is therefore fully synthetic and self-generated. Each topic has a materialized gold set under the frozen generator, and tests recompute the world and tasks from the recorded seed and compare all committed hashes rather than trusting a hand-authored denominator. Discoverability is heterogeneous: relevant items are partitioned across lexical, semantic, citation, reformulation and restricted/provider routes; two route labels deliberately share a backend; duplicate locators share content identity; revisions share identity but change digest; extraction-question-shift cases require legitimate rereads; and unavailable-route cases distinguish transport censoring from absence. Budgets are below exhaustive probing, so route allocation and stopping remain observable variables.

The completed suite has 390 tasks over five case families, drawn from 78 topics over 1,210 documents, frozen at suite fingerprint `2f6936ba52fb`. Every candidate uses the same host-owned session. The session alone can access the world, meters queries and reads, records route stops and transport states, and evaluates claims after the candidate returns. The candidate receives only a public task view and label-free retrieved records; protected gold and the relevance rule are never passed through the system interface.

External benchmark tasks. AutoResearchBench provides Wide and Deep scientific-discovery tasks [15]. The access audit establishes a narrower fact than an external result: Wide’s official exact IoU/recall/precision path scores normalized arXiv identifiers without a credential, while its sampled maximum-IoU-at- k family is noisy because upstream sampling is unseeded. Deep’s official title-matching evaluator needs an OpenAI-compatible judge. The Wide scorer was exercised on synthetic scorer input to verify that credential independence; no matched Wide result against a baseline is archived in this revision, and the scorer is not claimed to be vendored here.

SAGE is retained as neighbouring work and as a lexical-baseline warning [6], but is not executed as if its published setting were available: the audited repository lacks the 200,000-paper retrieval corpus described by the paper and lacks an official evaluator. Any substitute corpus or weighting rule would be a declared deviation rather than the SAGE benchmark. MetaSyn’s separation of retrieval from screening is adopted as a measurement constraint [14]; its retrieval/screening scoring path is credential-free by source audit, but no end-to-end result is archived here. AgentSLR remains a protocol-driven systematic-review neighbor/baseline family rather than an official evaluator dependency [10].

Frozen external retrieval snapshots and live-provider campaign. The prospective external design freezes a gold-blind candidate pool per query before comparing systems, then runs a smaller predeclared live subsample. Pool coverage must be reported as a shared ceiling because a gold-blind pool can omit gold. Live requests are designed to retain URL/parameters, query derivation, timestamp, raw response and typed transport outcome. The capture machinery and manifest exist and have been exercised, but the final result-bearing external pool and live campaign have

not been executed in this revision. Therefore no external cost, latency or recall result is inferred from the capture audit.

7.4 Systems under test

The implementation evaluated here is the ORION discovery layer; it is the artifact under test, and the rest of the paper calls it the *governed system* so that the claims read as claims about the mechanism rather than about a product. The frozen offline companion compares the governed system with six confirmatory comparators: a BM25 keyword retriever, a dense retriever, a sparse-dense hybrid, a one-pass retrieval-augmented pipeline, a single-route agentic searcher, and a protocol-driven systematic-review comparator. A seventh comparator, an adaptive multi-route exploratory searcher, was kept explicitly exploratory and was not retrofitted into the frozen confirmatory baseline set. The registry that binds each of these to its frozen system identifier is archived with the campaign.

These controlled-index systems are mechanism comparators, not claims to replicate a production BM25 engine, a particular embedding model or an external agent implementation. They consume the public synthetic route interface under identical budgets so that the variable under test is route/read/stopping governance. Their relevance screen sees only the public question and retrieved bibliographic text. The authored synthetic text makes screening deliberately easy; the experiment is not presented as evidence about classifier quality.

Six frozen ablations remove exactly one governance mechanism or authorize one forbidden transition: route-independence checking, the question coordinate in the read ledger, route/task stop separation, unavailable-route open state, coverage-diagnostic non-authority, and content-identity deduplication. All use the same session and budgets as the governed system.

7.5 Identity, reads and stopping

A read is keyed by content identity, content digest and extraction question in the full system. A new question can therefore reopen unchanged content for a legitimate reread, while mirrors with the same content identity/digest do not become new evidence. The corresponding ablations remove the question coordinate or replace content identity with document locator to expose suppression or duplicate work.

Route stop and task closure are separate actions. A route may stop after its public probe space is exhausted or after a typed transport outcome, but task closure is invalid if relevant material remains reachable within budget. An unavailable route stays censored and open rather than being converted to an empty result. Coverage diagnostics may inform a route policy but have no authority to certify global completeness.

7.6 Measurements

The offline evaluator reports denominator-valid complete-gold recall, precision, missed-gold count, route use, route/content overlap diagnostics, marginal relevant gain after the first route, premature task closure, censoring, duplicate processing and legitimate rereads, plus host-owned resource counters. Claims are normalized by content identity rather than locator so mirrors cannot inflate recall. Rich per-run artifacts retain route calls, reads, stop events, authority flags and raw-artifact hashes; the publication summary binds the full 16,380-record and rich-artifact sets by SHA-256.

The external measurement plan additionally defines official Wide/Deep scores, screening metrics and ranking-only metrics where their denominators and task semantics support them. Those definitions do not imply that an external result exists.

7.7 Statistics and authority

The statistical unit is the task. Repeated stochastic runs nest within task and system and never count as additional tasks. The frozen plan defines paired bootstrap comparisons, Wilson intervals for standalone binary rates, superiority/non-inferiority margins, a cost ceiling and Holm correction for the external confirmatory analysis. It also defines an authority ladder. The completed offline suite has $n = 390$, which reaches the precision tier the plan committed to in advance—registered in that plan as `TIER_B_committed` and requiring $n = 385$ —at an achieved half-width of 0.0496; because that half-width exceeds the frozen superiority margin, the primary offline result carries the plan’s mandatory underpowered label. The three deterministic repeat seeds verify stable execution; they do not manufacture additional tasks. We therefore report descriptive absolute differences and mechanism diagnostics, with no promoted superiority decision from the offline companion.

For a future external confirmatory result, the frozen primary rule requires the family superiority margin of 0.03 under matched resources with the premature closure safety guard no worse than 0.02. Non-inferiority uses the predeclared 0.01 margin and is inadmissible above the 1.25 consumed-cost ratio. Secondary families use Holm correction. These rules remain prospective because no external system result is reported here.

7.8 Failures, exclusions and contamination

Candidate failures, malformed outputs, budget exhaustion, abstention and typed transport failures remain outcomes. Provider unavailability cannot become evidence of absence. The frozen evaluator uses an explicit failure taxonomy and terminal precedence; the publication table therefore warns that a lower-priority mechanism can be masked by a higher-priority terminal classification such as premature closure. Public benchmark questions and gold identifiers also create a contamination risk for future live search. A credential-free search-time self-exposure diagnostic over the frozen 400-task AutoResearchBench Wide candidate run is archived with the external results and reports a task rate of 0.0. That diagnostic does not claim absence of indirect, training-time, or live-provider contamination.

7.9 Decision rule

The paper-level rule is fail-closed. The offline companion can support a mechanism claim only at descriptive authority. External discovery superiority requires an archived denominator-valid external result under the frozen matched resources and safety rule. If that evidence is absent, blocked or undetermined, the claim is narrowed rather than promoted by analogy from the synthetic world.

8 Results

8.1 Authority of the result

Two result-bearing evidence families are complete in this revision, with different authority. First, the frozen offline complete-gold companion contains 390 tasks, 14 systems (the governed system, six confirmatory baselines, one explicitly exploratory comparator, and six ablations), and three predeclared repeat seeds, for 16,380 normalized run records. The systems are deterministic; the repeats therefore test harness stability and do not increase the statistical unit beyond 390 tasks. The precision tier the frozen statistical plan committed to in advance requires 385 tasks, which the assembled 390 tasks reach at an achieved half-width of 0.0496 for a standalone rate. Because that

System	Recall	Premature closure	Pass	Cannot check	Fail
Governed system	0.979487	0.00	0.82	0.03	0.15
Protocol-driven SLR	0.666667	1.00	0.00	0.00	1.00
Sparse+dense hybrid	0.555556	1.00	0.00	0.00	1.00
One-pass RAG	0.555556	1.00	0.00	0.00	1.00
BM25/keyword	0.333333	1.00	0.00	0.00	1.00
Dense-route comparator	0.333333	1.00	0.00	0.00	1.00
Agentic single route	0.333333	1.00	0.00	0.00	1.00

Table 1: Controlled-index descriptive comparison. Rates are over 390 tasks; the failure column on the governed system’s row is honest budget exhaustion, not premature closure. The frozen plan’s underpowered label applies to the primary offline comparison, so no superiority interval or decision is reported here. A prospective external Wide campaign was executed but failed its frozen transport validity gate; its outcome is retained separately as undetermined.

half-width still exceeds the frozen superiority margin of 0.03, the plan’s underpowered label attaches to the primary offline result: no superiority decision from the offline companion is promoted to a paper-level claim.

Second, the credential-free MetaSyn probe is a bounded external retrieval/screening result. It completed all 86 released test reviews using the pinned MetaSyn BM25 title+abstract retriever followed by the predeclared public-protocol screening rule, and was scored by the pinned official ID-only evaluator. That probe is not the complete multi-provider system and is therefore reported separately from the paper’s still-open matched external comparison against a baseline.

The offline campaign is summarized by a single immutable publication record. It binds the full 16,380-record set by SHA-256 `27b8e55b68a65906fe0971ed2f24a814d31d66a1423d38782bb121c7f06e1525` and the corresponding rich-artifact hash list by `ed5cf7bd823fcabd6a57658a05560bd5f1e30256aa3f8eb8404325c493ed3cfe`. The exact run manifest was committed before the first outcome snapshot. The mechanism projection used for Figures 4–5 is archived separately; it records the same two source digests and is regenerated from host-owned traces rather than copied from manuscript numbers. The MetaSyn result and its exact integer error decomposition are bound by two further archived evaluator records. Section 13 names every one of these artifacts and where it lives.

8.2 Controlled-index comparison

The governed system attains mean complete-gold recall 0.979487 and precision 1.0. It passes 319 of 390 tasks, fails on 59 tasks after honestly exhausting its route-call budgets without certifying completeness, and leaves 12 tasks undetermined because a route holding unretrieved relevant material becomes unavailable; it never certifies a censored task complete. Its premature-task-closure rate is 0. The strongest of the six protocol-frozen confirmatory baselines is the protocol-driven systematic-review comparator, with mean recall 0.666667. That comparator claims task completion on all 390 tasks while reachable gold remains, so its premature-closure rate is 1.0. The descriptive recall difference is therefore +0.31282, with a descriptive premature-closure difference of -1.0. These numbers are evidence about the authored controlled world, not an estimate of real-web superiority.

The two systems are identical through their first three search calls in this authored world. The protocol-driven comparator then reaches mean recall 0.666667 by query 7 and remains at that value through query 12. The governed system is at 0.757835 by query 7 and continues to gain, reaching 0.979487 by query 12. Figure 3 therefore localizes the descriptive gap to evidence gained after the early common search prefix rather than to an initial retrieval advantage.

8.3 Mechanism ablations

The safety ablations fail in the directions the frozen design intended them to probe. Removing the route-independence check reduces mean recall to 0.444444 and produces premature closure on every task. Allowing a route stop to certify task closure yields recall 0.333333 with the same 100% premature-closure rate. Letting a coverage diagnostic acquire stopping authority yields recall 0.555556 and again closes every task prematurely. These are negative controls: their purpose is to demonstrate that the forbidden transitions are observable, not to serve as competitive systems.

Removing content-identity deduplication is less catastrophic but measurably wasteful in the controlled world: mean duplicate-processing rate rises from 0 to 0.101446, 145 of 390 tasks exhaust the read budget, pass rate falls from 0.817949 to 0.605128, and mean recall falls to 0.952707. Removing the question coordinate from the read ledger leaves aggregate recall unchanged at 0.979487 but changes the extraction-question-shift mechanism exactly where expected: the mean count of legitimate rereads on that family falls from 3.666667 to 1.0. This is a mechanism result, not evidence of a recall gain from the ledger. Finally, removing the unavailable-route open state leaves mean recall numerically unchanged but turns the 12 censored undetermined cases into asserted completions and premature-closure failures. The distinction is therefore one of epistemic safety rather than headline recall.

The lexical route contributes 3.0 first relevant identities per task on average. Semantic and citation routes each contribute 2.0 additional identities, reformulation contributes 0.848718, and the restricted route contributes 0.966667; the last mean is below one because tasks that end censored or budget-exhausted before the restricted route is consulted contribute zero. Contribution and independence are deliberately separate variables: reformulation adds relevant content but shares the lexical backend and therefore cannot be promoted to an independent capture occasion.

Figure 5 exposes why numerical disjointness is not equated with route independence. In particular, lexical and reformulation have zero mean content overlap here yet are still rejected as independent because they share a backend. Conversely, lexical and semantic have nonzero mean overlap (0.07557) but remain structurally independent because neither backend nor query derivation is shared. The independence decision is therefore made from capture construction, while overlap remains a measured diagnostic.

8.4 MetaSyn external retrieval and screening probe

The corrected MetaSyn probe completed all 86 released test reviews with no failed review. Under the pinned official ID-only evaluator, macro retrieval recall is 0.7485 and retrieval precision 0.0627. Conditional retention among retrieved reference items is 0.7224; final inclusion recall is 0.5651, inclusion precision 0.1426 and inclusion F1 0.2033. The evaluator reports post-retrieval loss 0.1834 and screening accuracy 0.7228. Mean candidate output is fixed at 200 retrieved and 60 included items per review by the declared probe budget.

The evaluator-side integer ledger makes the stage loss concrete. Across the 86 reviews there are 1,677 reference-article links. The top-200 retrieval pools contain 1,079 of those links, so 598 are retrieval false negatives. Of the 1,079 retrieved reference links, 736 survive the deterministic screen and 343 are lost post retrieval. The final included sets therefore miss 941 reference links in total. Gold is loaded only on the evaluator side after candidate output is frozen, and fresh regeneration verifies every deterministic official per-review metric before emitting the counts. These results identify where the bounded probe fails; they do not establish superiority for the full system.

8.5 Failure analysis and external status

Under the evaluator’s fixed terminal-failure precedence, premature closure is the dominant failure class in the controlled companion; the exploratory adaptive comparator also exposes 65 transport-failure classifications, while the no-dedup ablation exposes 145 budget-exhaustion failures. The governed system’s only terminal failure classification is honest budget exhaustion on 59 tasks: 319 tasks pass, 12 are censored, and premature closure remains at zero. The full per-system taxonomy is archived beside the publication summary.

A gold-defined route-stop replay is archived as supplementary material. The governed system makes 13 route-stop false positives in 1,950 route-stop events (0.006667): those restricted-route cases still have a theoretically reachable gold identity when the route stops. These do not become task-level false closures. The unavailable-route obligation remains open, so the governed system leaves the task undetermined rather than asserting completeness. Among the 1,878 governed task-routes that reach the oracle exhaustion point, there are zero route-stop false negatives. The restricted route records 377 post-exhaustion attempts in total; no task takes the two or more further attempts required for a route-stop false negative. This null-on-FN result is reported rather than being silently omitted, and it demonstrates why route-stop error and task-stop safety must be measured separately.

The remaining external authorities are kept separate from the completed MetaSyn probe. AutoResearchBench Wide’s released scorer is deterministic and credential-free, while the official Deep metric uses the released LLM title judge. That judge has now executed for the declared keyless public-arXiv probe: all 600 released Deep tasks were scored under the pinned judge prompt and model reached through a local protocol adapter, and the judge matched the reference title on 0 of them, a hit rate of 0.000. A nine-case positive/negative judge control—verbatim, truncated, and case-variant matches alongside distinct-paper non-matches—passed in full before the archive was accepted, so a zero-match outcome is attributable to the probe rather than to an always-no-match judge failure. This is a bounded external probe, not a matched Deep comparison against a baseline, and no Deep superiority claim is drawn from it. SAGE still lacks both the published retrieval corpus and an official evaluator. The frozen three-route live campaign also remains undetermined: its arXiv route is credential-free, but its two OpenAlex-backed route families require a funded OpenAlex API key, which the verified same-repository preflight did not find. No absent credential is converted into a negative outcome or a zero-cost measurement.

That zero was then diagnosed rather than left standing. A judge-free stage attribution over the same archive compared all 10,850 returned candidate titles against the 564 tasks that carried both a reference and candidates: exact-title recovery 0, substring recovery 0, and token-overlap recovery at a half-token threshold on 8 of them. Because a plain string matcher recovers almost nothing from the candidate sets themselves, the failing stage is candidate *generation* and not scoring, judging, or reference formatting. The attribution names a specific mechanism: questions whose incidental document-apparatus vocabulary—*supplementary*, *appendix*, *figure*—is treated by the acquisition mechanic as content vocabulary, so a title such as *Supplementary Orbit* outranks the paper the requested number actually sits in.

A successor acquisition mechanic was then built against that named mechanism and validated under a freeze written before any corpus was generated. The successor, a discriminative term-gating mechanic, uses index statistics only—never the gold set, the target, or the task family—discarding question terms whose document frequency exceeds five percent of the corpus, ranking the survivors by inverse document frequency at the baseline’s own query width, and demoting rather than deleting documents that agree on fewer than two discriminative terms. On an authored corpus of 2,760 documents in which the named mechanism is present by construction and the denominator

is complete, the current mechanic recovers the target for none of the 120 echo tasks within the top ten while the successor recovers all of them, at a rank-one recovery rate of 0.983. The paired McNemar test over those tasks is discordant on 120 pairs, every one of them favouring the successor. The harm guard is not vacuous: the current mechanic already scores 1.000 on the 60 tasks written without echo, and the successor does not damage them. BM25 over the same query recovers much of the top-ten rate (0.633) but almost none of the rank-one rate (0.0083), so the query-side gate rather than document-side weighting is what carries rank one. A pre-registered specificity control behaves as required: on 40 tasks whose question and target share no content token, the successor gains exactly 0.000, confirming the arena isolates the lexical mechanism instead of rewarding the successor generally.

This result is scoped to constructed reproduction only, and that scope was fixed before any number existed. The official probe cannot be re-run here: its frozen input is not in the repository and both public providers refuse this environment’s proxy. The successor is therefore evidence that the mechanism the attribution names is by itself sufficient to drive a lexical candidate generator to near-zero recovery, and that a discriminativeness-grounded generator is not driven to zero by it. It licenses no statement about target-hit rates on AutoResearchBench Deep, does not revive the archived probe, and does not discharge the paper’s successor-mechanic terminal, whose arena must be external and whose evaluator custody must sit outside the lane that authored the mechanic. One further limitation is declared in advance rather than discovered later: the successor requires corpus-level document frequencies, which are exact offline but against a live provider need a result-count endpoint or sampled index that this study does not supply.

A separate gold-blind OpenAIRE identity discriminator first established that structured arXiv persistent-identifier objects were available on 3 of 12 frozen public Wide questions, with 12 of 12 HTTP requests successful; that discriminator was forbidden to score or promote the paper. A subsequently frozen 400-row OpenAIRE/Crossref matched campaign then issued exactly 1,200 logical provider requests over one shared capture. The 400 Crossref discovery calls and 400 direct OpenAIRE calls succeeded, but every predeclared OpenAIRE DOI-crosswalk call returned HTTP 400. The observed provider-success fraction was therefore 0.666667, below the frozen 0.90 validity gate. The balanced baseline, direct-first diagnostic and governed projection were byte-identical, and their paired IoU differences were zero on all 399 distinct questions. The frozen analyzer correctly emitted an undetermined terminal, not a valid negative result.

Two further prospectively frozen campaigns followed, and together they correct the natural reading of that first one. A second campaign changed only the crosswalk encoding, from a single quoted OR expression to repeated persistent identifier parameters; its gold-blind transport probe returned HTTP 200 with an empty result array, below its prospectively frozen minimum of one, so acquisition never began and no score was produced. A third campaign changed the encoding again, to the documented identifier-filter OR list, and this time the transport probe validated: its receipt and raw response hashes were bound into the capture manifest and independently revalidated at score time. All 400 rows were acquired and scored. Provider validity reached 0.715833, again below the frozen 0.90, so the analyzer again emitted an undetermined terminal rather than a negative result.

The correction is this. The three systems were byte-identical in the first campaign, and their evaluator outputs were byte-identical again in the third, where the cross-backend route *was* valid and the three candidate files were genuinely distinct. So the identity is not explained by the failed crosswalk. It is explained by the measurement floor. The frozen decision rule requires an average-IoU difference of 0.03 before superiority is supported, and the best-scoring arm in both completed campaigns achieved an average IoU of 0.003924 and 0.004374 respectively — roughly a seventh of the difference the campaign exists to detect. Deleting a competing arm outright would not reach

the threshold. Three systems that all retrieve almost nothing tie on nearly every question, so their aggregate scores coincide and their scored outputs are identical files.

That matters for how the accompanying interval should be read. Both campaigns report a paired bootstrap interval of $[0.0, 0.0]$ over 399 matched questions, with 399 ties and no wins or losses, from ten thousand resamples, at the same confidence level and percentile method the campaign froze. Taken at face value that is the strongest equivalence evidence a bootstrap can produce. It is not evidence of equivalence at all: resampling 399 identical differences returns a zero-width interval at any sample size, and the interval would appear exactly this conclusive with four questions rather than four hundred. We therefore report no equivalence or non-inferiority conclusion from either campaign. Three reporting defects are recorded with them: both artifacts publish the unseeded sampled maximum-IoU-at- k family that this paper’s own scorer excludes by rule, that family decreases in k — which a maximum over the top k cannot do — and every runtime total is reported as exactly zero, which is untransported measurement rather than measured cost.

The terminals, workflow receipts and expiring Actions archives are retained in the repository, with checksummed mirrors beside them and a separate resolution audit; Section 13 names each location. A further campaign would require a new prospective freeze; none of these captures may be reinterpreted after the fact. The prerequisite is no longer a transport encoding. It is finding out why retrieval scores near zero on a benchmark whose gold is reachable, because a fourth transport version scored at the same floor would spend the same effort to learn the same nothing.

Therefore the externally supported discovery-benefit claim for the full system remains undetermined. The controlled-index result supports the narrower statement that the frozen governance mechanisms behave as designed under a complete denominator and matched host-owned budgets; the MetaSyn result supports only its declared external retrieval/screening probe. Bounded AutoResearchBench probe results, when archived, must remain separately labeled unless they exercise the final matched system under the frozen promotion plan.

9 Post-saturation unresolved-route validation

The post-saturation literature search made a wider systems claim testable only after granting more prior art to the acquisition layer. Verification-aware claim coverage, decision-sufficiency and utility stopping, coverage-driven deep-research completion, finite source-obligation certificates, and generic workflow-obligation graphs are all treated as donor mechanisms rather than as novelty claimed here [11, 2, 7, 1, 17]. The remaining question is narrower: after those mechanisms have done everything they can on the routes that are actually available, what scientific authority exists to close the task when a *material* route is unavailable, censored, provider-invalid, or still carries an unresolved question-conditioned processing obligation?

The exact-contract battery. The battery freezes five acquisition families—scientific literature, web/API evidence, dataset registries, code/artifact repositories, and experiment/measurement routes—with eight hostile archetypes per family. The cases distinguish all-obligations-discharged closure, one available route remaining open, unavailable/censored/provider-invalid material routes, duplicate route coverage, already-acquired but not question-processed content, and low-utility hard routes that stay open. Route-local decisions are stop, continue, or undetermined; task terminals are close, continue, or undetermined, with undetermined reported as its own outcome rather than collapsed into a refusal to close. Route order is seed-permuted, and held-out variants add unavailable *non-material* decoy routes so that blanket “any unavailable source blocks” logic cannot pass.

Strong products. The primary comparator, B1, is deliberately donor-complete. It receives route identity/materiality, HALT-like evidence coverage, decision sufficiency, utility stopping, finite route certificates, deduplication, route state, and the question-conditioned processing coordinate. It also makes the same correct route-local decisions as the authority rule under test. Its only systematic difference is global aggregation: once all *available* material routes locally stop and their acquired content is processed, it treats the task as closed, excluding unavailable/censored/provider-invalid material routes from the closure denominator. A second comparator, B2, allows a global sufficiency signal to authorize task stop. A third, B3, is an ideal information-equivalent product given the exact unresolved-route semantics of the authority rule and is expected to tie; it is an expressivity boundary rather than a weak baseline.

Protected result. On 400 prospectively frozen protected cases, the authority rule obtains 400/400 exact scientific closure decisions ($\text{ESCD} = 1.000$), versus 250/400 (0.625) for B1 and 100/400 (0.250) for B2. The paired advantage over B1 is +0.375 with a predeclared domain-stratified bootstrap 95% interval $[0.3275, 0.4225]$, above the registered +0.10 practical margin; exact McNemar counts are $b = 150, c = 0$. The authority rule records zero false task closures, preserves 1.000 clean task-stop accuracy when every material obligation is discharged, and has a positive advantage over B1 in all five acquisition families. B1’s 150 closure failures are exactly the three unresolved-material-route archetypes—unavailable, censored, and provider-invalid—while it passes every other archetype.

An independent implementation reproduces all 400 case identities, protected-bundle and canonical decision-row hashes, every headline arm count, and zero B3 decision mismatches. The non-material unavailable decoys cause no materiality error, and all statistics include the held-out decoy variants.

Boundary result. B3 obtains 400/400 and is decision-identical to the authority rule. Thus the result does not establish inherent expressivity, centralization superiority, or ownership of stopping/obligation machinery. It supports a bounded authority distinction: with realistic donor-complete interfaces, an unavailable/censored/provider-invalid *material* acquisition route must stay unresolved rather than disappearing from the task-closure denominator. An ideal product supplied that same rule is equivalent.

The result also does not repair the historical external Wide campaign. The OpenAIRE/Crossref campaign is still provider-invalid and undetermined; no independently implemented retrieval engines or new live multi-provider campaign were executed in the battery. Deployed and retrieval-engine generality therefore stays undetermined.

10 Successor interface: structure-conditioned and source-grounded acquisition

The acquisition–closure separation also applies when a search route is generated from a structural need rather than topical vocabulary. A successor route may derive its query from a method signature, a failure signature, a representation analogy, or an unresolved atom study. This remains an acquisition choice. The label “structural” does not make a route independent: backend identity, query-derivation identity and capture identity must still earn independence under the accounting used elsewhere in this paper.

Returned material is retained as a candidate donor, not a transfer certificate. Before a structurally interesting candidate is passed downstream, its assumptions, protected invariants, effects

and reconstruction obligations can expose an acquisition-local obstruction or unresolved mapping. Semantic distance is not evidence of novelty, and surface similarity is not evidence of valid transfer. Whether a correspondence actually holds, and whether a candidate deserves scientific promotion, are separate problems that this paper does not decide.

Two source-grounded historical substrates are available to this acquisition layer, with intentionally limited authority. A discovery morphology atlas freezes a 51-episode, four-domain, ten-coordinate descriptive basis with explicit no-Jump controls and two post-basis no-material-change rounds. A failure atlas freezes a 30-episode revision-aware schema. In both cases, historical material remains mechanism-extraction and benchmark-design evidence: private cognitive steps stay non-inferable, competing interpretations are retained, and curation does not transform famous examples into protected confirmatory gold. The stronger corpus-ready state of the failure atlas was not claimed because independent domain-expert agreement was not obtained.

The prospective atom results reinforce the same boundary. Broad tension, imagination and concept labels decomposed into multiple candidate atoms rather than becoming new search-route authorities, and the zero-error Jump candidate tied the verified representation-regime parent on both frozen splits. None of those outcomes authorizes an unresolved route to be closed, corpus completeness to be inferred, or a historical analogy to be promoted into scientific evidence.

Route-local stopping therefore remains local under this extension. Exhaustion, censorship, provider failure, budget stopping, a temporarily empty structural route, or a stable descriptive atlas does not establish that no useful source or method exists. A route stays open or revisitable whenever a new representation, source family, or upstream structural object could materially change the search. The successor interface extends the ways candidates can be acquired without weakening the central rule: acquisition evidence does not acquire scientific-closure authority merely because it came from a richer representation or a curated scientific history.

11 Related work

AutoResearchBench establishes that both specific-target and open-ended scientific-literature discovery remain difficult for current agents [15]. The acquisition frontier is already strong and heterogeneous: SAGE shows that tuned lexical retrieval can outperform LLM retrievers inside deep-research workflows [6]; AgentIR uses the agent’s reasoning trace as a retrieval signal [3]; SIEVE combines fielded Boolean admission, structure-rich inspection and selective section fetching [13]; and breadth-first bibliography expansion recovers material missed by API-only retrieval [12]. Rather than reserve a weaker baseline, Definition 1 places all runnable instances and their outcome-blind functional analogues inside the donor-complete policy class. Their actions enlarge the acquisition envelope; a candidate contribution must survive that enlargement.

Screening and stage attribution form a second neighbourhood. MetaSyn separates retrieval from eligibility screening and shows why a retrieval ceiling does not imply an inclusion-recall ceiling [14], and AgentSLR shows that end-to-end systematic-review automation is an active baseline family in its own right [10]. This paper adopts their measurement discipline—stage attribution and denominator caution—and inherits their warning that one aggregate recall number hides where the evidence was actually lost.

The stopping and obligation literature owns more than “when to stop.” DeepControl and decision-theoretic screening optimize continuation against information value, cost and payoff [16, 5]; HALT uses claim–evidence coverage and abstains when it cannot verify that coverage [11]; MiCP controls adaptive coverage [18]; and structured Search-R1 learns sufficiency and gap judgements [8]. Don’t Stop Early couples coverage-driven objectives to evidence-aware completion [7], while

Confidence-Based Stopping targets decision sufficiency rather than recall alone [1]. KnowPlan provides finite source-obligation certificates [2]; iCORE binds work and obligation transitions to evidence [17]; and Science of Intent explicitly studies closure gaps and misclosure [4]. The present theory grants all of these mechanisms to the donor state. Its residual is the factorization question in Theorem 2: after composition, does the retained state actually distinguish every closable world from every materially open world? If yes, the product is sufficient and may tie. If not, no new classifier over the same state can manufacture authority.

Question-conditioned memory has its own line of work: MemChain constructs question-conditioned evidence plans and grounded memory traces [9]. The envelope treats memory as an optionality-bearing scientific state, not merely a token cache. Content identity prevents duplicate rediscovery from inflating coverage, whereas question-conditioned processing state permits a changed extraction frame to reopen already encountered material. Whether that additional coordinate matters is decided by fibre separation and prospectively hidden future questions, not by claiming generic memory novelty.

12 Limitations and integrity

The completed controlled experiment is deliberately synthetic. Its complete denominator is what makes missed relevant material and premature closure observable, but the authored routes, public probe vocabulary and easy screening text make it a mechanism test rather than a realistic estimate of retrieval difficulty. The observed +0.31282 recall difference therefore cannot be transferred to the open web or to AutoResearchBench by analogy.

The contribution is a comparison and composition theory rather than a claim to own its components. Scientific retrieval-augmented generation, agentic literature search, automated systematic review, capture-recapture estimation, lexical, dense, reasoning-aware, field-aware and citation-expansion retrieval, learned or decision-theoretic stopping, conformal coverage, finite obligation ledgers and question-conditioned memory are prior art and enter the donor class. The claimed theoretical residual is the joint acquisition-authority envelope, its run-conditional acquisition ceiling, the closure-factorization criterion, the indistinguishable-world lower bound and the maximally permissive sound completion relative to an explicit obligation contract.

The suite carries 390 tasks, which reaches the plan’s committed precision tier at an achieved half-width of 0.0496 for a standalone rate. The frozen statistics plan fixed those tiers before outcomes were inspected and assigns this campaign that committed authority; because the achieved half-width still exceeds the frozen superiority margin of 0.03, the primary offline result carries the plan’s underpowered label whatever the task count becomes. All 16,380 system/task/repeat executions reproduce exactly, and the 3 deterministic repeat seeds demonstrate stable execution; repeats nest within task and never increase the statistical unit.

The exact-contract battery broadens the *contract families* tested, not the realism of retrieval. Its 400 cases span five acquisition settings and include seed-permuted route order plus non-material unavailable decoys, but route state, route materiality and question-processing state are explicitly supplied rather than inferred from noisy web behavior. The 1.000 exact-closure ceiling therefore supports the fail-closed authority rule only on the registered exact contracts. No independently implemented lexical/dense/reasoning retrievers or new live provider campaign were used to establish retrieval-engine generality, and the ideal-product arm shows that an ideal product given identical unresolved-route semantics is extensionally equivalent.

Open-world completeness remains generally unprovable from finite search. External providers are mutable, rate limited and differently scoped, and public benchmark questions and gold iden-

tifiers create search-time contamination risk. The corrected MetaSyn run is external but narrow: it uses pinned BM25 retrieval plus a deterministic public-protocol screening rule and the official ID-only evaluator, not the complete route system. Its exact evaluator-side ledger contains 1,677 reference links, of which 598 are missed at retrieval and another 343 are lost after retrieval, leaving 941 final false negatives. These numbers characterize that probe and are not promoted to a general claim about the system.

The theory does not remove that impossibility. The maximality theorem is relative to a declared obligation/realizability contract; it cannot prove that a finite open-web route registry contains every material route. The acquisition ceiling is conditional on the captured run and on a valid denominator, not an estimate of the best route that might be invented later. The envelope itself depends on the task distribution, budget, access interface and donor class, so cross-domain superiority is an empirical hypothesis rather than a corollary of the formalism.

Several external authorities remain partially unavailable rather than negative. AutoResearch-Bench Deep official scoring has now executed through a local OpenAI-compatible adapter for the bounded keyless probe, with the judge control passing in full, but the matched multi-provider Deep comparison remains unexecuted. SAGE still lacks its published retrieval corpus and official evaluator. The separately archived OpenAIRE/Crossref Wide campaign failed its prospectively frozen provider-validity gate and remains undetermined; its observed zero paired effect is not a valid null or negative result. The exact-contract battery does not repair or supersede that external failure. Provider or credential unavailability is likewise kept undetermined, never converted into evidence of absence or a flattering zero-cost result.

The 768-cluster successor also remains entirely prospective. Its arithmetic freeze establishes neither that the task worlds exist in compliant form nor that the planned effect is real. Four source-disjoint arena snapshots, reproduced baseline worlds, a capability-matched external comparator, live response/cost receipts and independently separated evaluator/gold custody are still missing. No part of that protocol is counted as external evidence in the present paper.

The mechanism ablations themselves impose an interpretive limit. Removing the question coordinate from the read ledger changes legitimate reread behavior but does not change aggregate recall in this suite; removing the unavailable-route open state changes censoring/closure semantics while leaving aggregate recall numerically unchanged. These null-on-recall mechanism findings are retained rather than converted into a stronger claim.

All result-bearing statements in the abstract, Results and conclusion are mapped to immutable archived artifacts through the claim ledger, the current non-promotion ledger, and the exact-contract result and verification receipts; the artifact names and paths are listed in Data and code availability. Rows without an archived supporting result stay undetermined; an access audit, implementation test, or credential preflight cannot substitute for a system outcome.

13 Data and code availability

The implementation repository contains the controlled index, host-owned evaluation harness, the frozen offline systems and ablations, the descriptive analysis code, and the publication-layer regeneration commands. Every artifact named in the narrative is listed here with its path, so that no repository location has to be carried in the prose.

Preregistration

The study was preregistered and frozen before any outcome was accessed, under the identifier `P2.open-world-discovery.v1`. A reader checking whether a reported analysis was planned or

chosen after the fact should compare against that freeze; it is the reason every number below can be traced to a decision made in advance of seeing it.

Frozen design and measurement rules

The design freeze is `protocol/PROTOCOL_V1.json`; the measurement and statistical rules are `protocol/MEASUREMENT_PLAN_V1.md` and `protocol/STATISTICAL_PLAN_V1.json`; external-artifact access states are `protocol/EXTERNAL_ACCESS_AUDIT_V1.json`; and the prospective execution binding for the result-bearing offline campaign, including the exact subject revision and Git blob identities, is `protocol/OFFLINE_RUN_MANIFEST_V1.json`, which also binds the frozen identifier of every system under test. The identifiers are `orion_full` for the governed system; `bm25_keyword`, `dense_retrieval`, `sparse_dense_hybrid`, `one_pass_rag`, `agentic_single_route` and `protocol_driven_systematic_review` for the six confirmatory comparators; and `adaptive_multiroute_exploratory` for the explicitly exploratory one. Each is a key of the systems map in `evidence/offline_results/RESULTS_SUMMARY_V1.json`, alongside the six ablation identifiers. The mapping from every result-bearing sentence in this manuscript to the archived artifact key that supports it is `protocol/CLAIM_LEDGER_V1.json`, with a human-readable view at `evidence/CLAIM_LEDGER_V1.md`.

The definitions in Section 5 are implemented in `src/orion/knowledge/`: work identity and read state in `identity.py` and `ledger.py`, route independence and coverage in `routes.py`, and route control in `route_control.py`.

Offline complete-gold companion

The frozen complete-gold world is distributed under `evidence/offline_gold/`; it is fully synthetic, regenerable from the recorded seed, and verified on every clean CI run against its committed content hashes and suite fingerprint.

The compact publication result is `evidence/offline_results/RESULTS_SUMMARY_V1.json`, and the mechanism projection behind Figures 4–5 is `evidence/offline_results/OFFLINE_MECHANISMS_V1.json`. The full per-system failure taxonomy is `evidence/offline_results/TABLE_P2-3_failure_taxonomy.md` and the gold-defined route-stop replay is `evidence/offline_results/TABLE_P2-S1_route_stop_oracle.md`. The command `python3 papers/paper-02-open-world-scientific-discovery/scripts/run_offline_companion.py --check` reconstructs all 16,380 runs and refuses drift. With `--write-raw DIR` it also emits the complete normalized JSONL result set and all rich run artifacts. The committed summary records SHA-256 digests of both complete sets, so a reproduction can be checked without requiring hundreds of generated files in the repository.

External benchmark artifacts

External benchmark artifacts are not redistributed. Their licences, access states and pinned revisions are recorded per artifact in `protocol/EXTERNAL_ACCESS_AUDIT_V1.json`, with source checks retained under `evidence/access/` and summarized in `protocol/TABLE_P2-1_freeze_manifest.md`. AutoResearchBench is pinned to upstream commit `a46c9bfb8968786f73f0a6a5b365b5384cd0f96d`; the released 1,000-record bundle is split by its explicit task-type field into 600 Deep and 400 Wide records before candidate custody. The Wide scorer is deterministic and credential-free. Deep official scoring uses the released title-identity judge and still requires an OpenAI-compatible endpoint; the repository workflow now records this authority precondition without exposing credential values. The

credential-free search-time self-exposure diagnostic over the frozen Wide candidate run is `evidence/external_results/AUTORESEARCHBENCH_SEARCH_CONTAMINATION_V1.json`.

The completed MetaSyn probe is archived as `evidence/external_results/METASYN_ID_0_NLY_PROBE_V1.json`. It binds all 86 released test reviews, the pinned upstream revision, dataset revision, candidate-output tree and official ID-only evaluator output. The evaluator-side integer error ledger is separately archived as `METASYN_SCREENING_FN_LEDGER_V1.json`; its regeneration reloads the pinned gold only after candidate output is frozen and verifies the official per-review metrics before emitting exact retrieval and screening false-negative counts. These artifacts make the bounded MetaSyn retrieval/screening result reproducible without reclassifying it as a result for the full system.

All three OpenAIRE/Crossref Wide campaigns are preserved rather than left behind an expiring Actions URL. Their exact results and run receipts are stored under `evidence/external_results/`, one prospective freeze per campaign under `protocol/`; checksummed mirrors of the shared public candidate capture, the failed first evaluator handoff, and the successful evaluator-only repair are under `evidence/ci_mirror/`. The mirror manifest records both archive-level and per-file SHA-256 digests. Evaluator gold never entered any candidate capture. The first campaign is undetermined because 400 of 1,200 planned provider calls failed; the second because its transport probe returned an empty result array below its frozen minimum, so acquisition never began; the third because provider validity reached 0.716 against a frozen 0.90. None is undetermined because evidence was discarded.

The prospective freeze for the successor acquisition mechanic is `protocol/P2_LEXICAL_ECHO_SUCCESSOR_FREEZE_2026-08-21.json`, written before any corpus was generated; the mechanic itself is registered there under the identifier `D4_DISCRIMINATIVE_TERM_GATING`.

The resolution audit that establishes the measurement-floor finding is under `evidence/audit/`. It reads the published campaign results and reports whether each could have produced a different verdict; it is reproducible from the repository and carries, in the same artifact, a control campaign with resolution so that a check firing on every input can be told from a finding.

The live-provider capture implementation and campaign manifest retain request URL/parameters, query derivation, timestamp, raw body, hash and typed transport status. The frozen live protocol requires arXiv plus two OpenAlex-backed route families. A non-leaking Actions preflight is archived as `evidence/access/LIVE_PROVIDER_PREFLIGHT_2026-08-16.json`; it confirms that arXiv is credential-free but that no funded OpenAlex API key is available in the same-repository execution context. Therefore the three-route live campaign cannot be frozen for execution here, and no raw live campaign or live monetary-cost ledger is fabricated. Operator re-scope 2026-08-17 removed the empirical-cost requirement: the sanctioned cost axis is the host-recorded search-query count (Figure 3, `protocol/TABLE_P2-2_baseline_ablation_cost.md`), not a live OpenAlex or arXiv API bill.

14 Reproducibility

Four levels are distinguished, and we claim only what the current artifacts support.

Fully reproducible without network access. The 390-task offline controlled-index companion regenerates from committed data and subject code. A clean GitHub Actions job independently rebuilds the 16,380-run publication projection and compares it to the committed summary. The frozen plan awards the result its committed precision tier together with a mandatory underpowered label on the primary offline comparison; reproduction does not elevate that authority.

Reproducible bounded external evidence. The pinned MetaSyn workflow reconstructs the

public/gold split, runs the credential-free BM25 plus public-protocol screening candidate, executes the official ID-only evaluator, and can regenerate the exact false-negative ledger. This is external evidence for that declared probe only.

Runnable credential-free benchmark lanes. AutoResearchBench Wide scoring is deterministic once the gold-blind candidate output is frozen. The same public-arXiv candidate machinery also supports the declared deterministic Deep exact-ID deviation. The official Deep title judge remains a distinct authority and is not replaced by the ID deviation when no judge endpoint is available.

Executed official Deep judge archive. The bounded keyless public-arXiv probe of the official Deep LLM title judge is archived as `evidence/external_results/DEEP_OFFICIAL_ARCHIVE_V1.json` with its nine-case positive/negative judge control in `DEEP_JUDGE_CONTROL_2026-08-17.json`. The checkpointed judge runner and the archive finalizer are both reproducible from the repository as `scripts/judge_checkpointed.py` followed by `scripts/finalize_deep_scoring.py`; the archive records the judge model, the pinned prompt digest, the run manifest hashes, and the bounded probe scope label.

Capture-capable but blocked as a final live campaign. The committed live layer is designed to archive mutable provider evidence rather than re-derive it, but the frozen route set requires funded OpenAlex access that is absent in the verified execution context. Until that authority is supplied and the campaign is actually run, raw live-provider outcomes, empirical live monetary cost and wall-clock, and the matched external discovery claim all stay undetermined. Query-count metrics from the offline companion are the completed cost evidence in this revision.

15 Ethics

This work does not collect new human-subject records or personally identifiable information. The frozen complete-gold companion is fully synthetic and regenerable from a recorded seed. External probes use publicly released benchmark questions and evaluator interfaces; third-party corpora are not redistributed where the access-audit table records a licence block or partial redistribution. Dual-use risk is that of any scientific-literature retrieval system: the paper evaluates discovery governance and does not present a deployment, surveillance, or human-facing study.

16 Conclusion

Scientific discovery systems should be compared as joint acquisition–authority controllers, not as a ranker followed by an unexamined stop decision. The envelope formalism makes that comparison donor-monotone and non-compensatory: stronger retrieval, reasoning, coverage, stopping, obligation and memory methods are absorbed into the comparator class, while false closure cannot be traded for report quality or lower cost. The acquisition-ceiling theorem attributes failures before proposing a repair; the factorization theorem states exactly when a product has enough state to implement scientific closure; the indistinguishable-world bound proves when new acquisition or new assumptions are unavoidable; and the obligation completion is maximally permissive among sound rules under its declared contract.

The evidence is deliberately heterogeneous rather than averaged into a single claim. The controlled-index campaign supports mechanism behavior but remains underpowered for its frozen superiority margin. The external 7/229 exposure result is an adverse candidate-generation diagnosis, and the provider-invalid matched campaign remains undetermined. The exact-contract battery

supports the predicted authority distinction—400/400 against 250/400 for a product that drops unresolved material routes—while the information-equivalent product’s 400/400 tie confirms that the residual is state sufficiency, not inherent centralized expressivity. These are components of a wider theory, not evidence of deployed general superiority.

The upward research target is now precise rather than merely narrower: exceed the guarded donor-complete envelope on source-disjoint naturalistic scientific worlds by inventing routes that lift the acquisition ceiling, preserving decision-relevant future options, and closing safely under independently held obligation custody. The frozen 768-cluster, four-arena successor is designed to test that claim, but it has not run and cannot change the current terminal.

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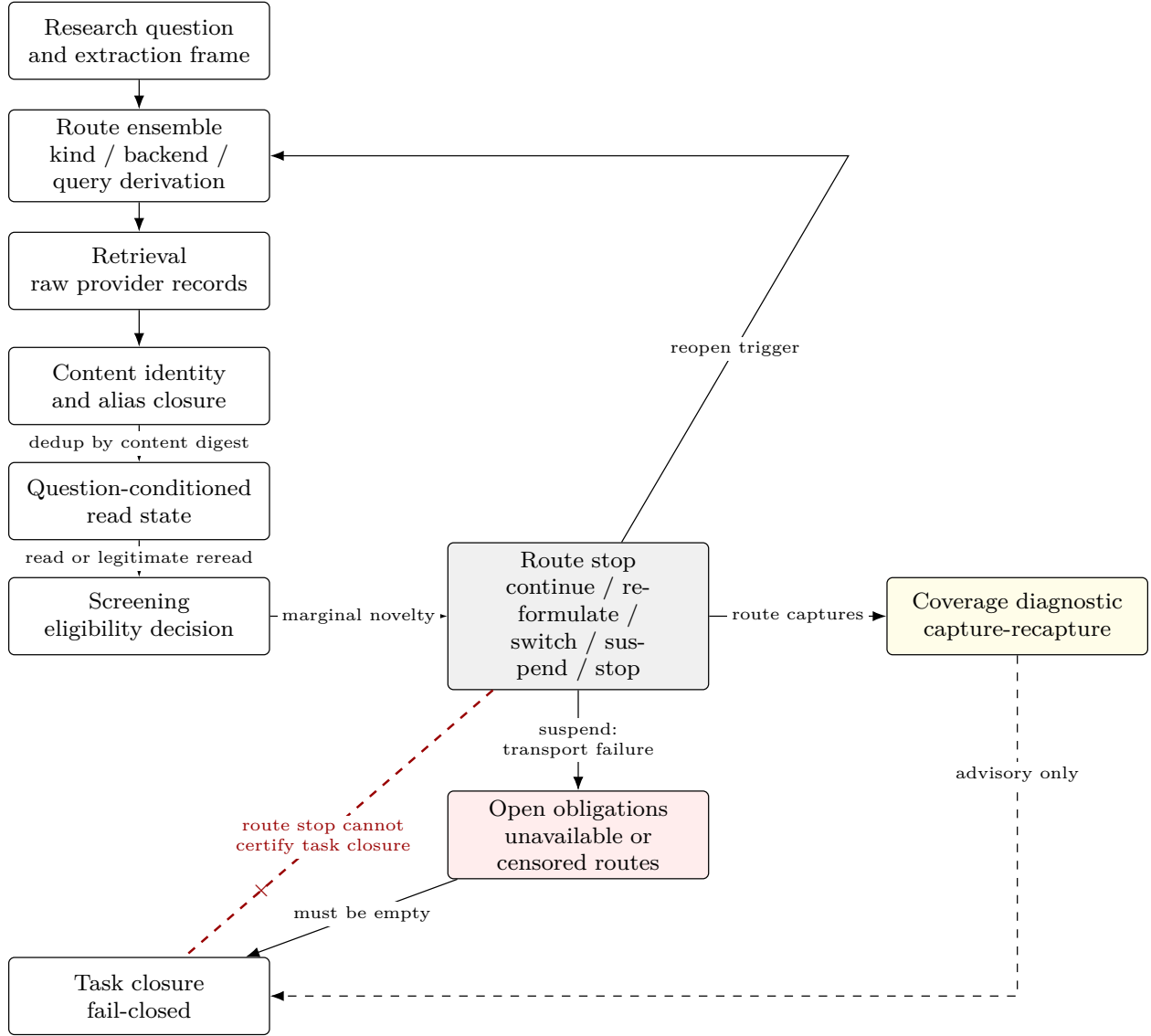


Figure 2: **Figure P2-1.** The discovery pipeline, separating retrieval, screening, route stopping and task closure. Solid arrows are authority-bearing transitions; the dashed arrow marks the coverage diagnostic, which informs but never certifies closure; the crossed arrow marks the transition the design forbids, namely a route stop certifying task-level completeness (Proposition 2). A route suspended by transport failure leaves an open obligation that closure must clear, so provider unavailability cannot become evidence of absence. The figure is generated from the same specification the tests read, so it cannot drift from the state machine it illustrates.

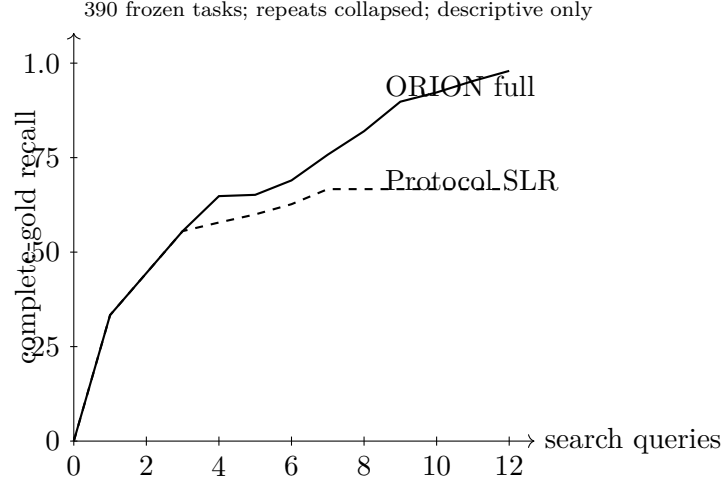


Figure 3: **Figure P2-2.** Cumulative complete-gold recall versus the host-recorded number of search queries for the governed system and the strongest frozen confirmatory baseline. Deterministic repeats are first checked for identical traces and then collapsed within task; the curves are means over the 390 tasks and are descriptive only. Synthetic wall-clock is intentionally fixed to zero, so the controlled companion supports the query axis, not an empirical-time claim.

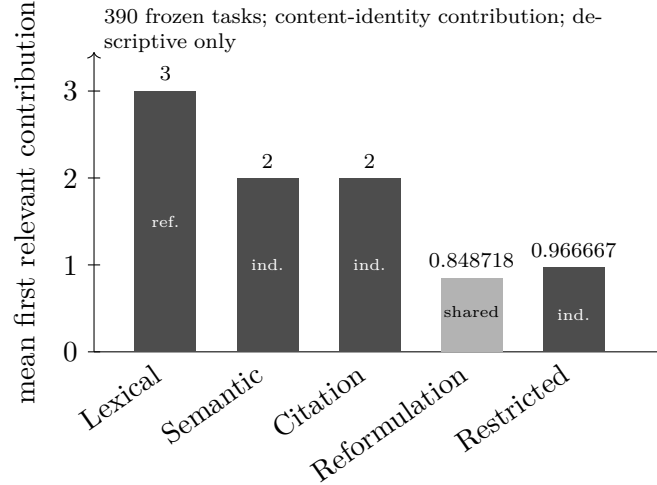


Figure 4: **Figure P2-4.** Mean number of first relevant content identities contributed by each route in the governed system’s campaign order. “Independent” is an earned structural annotation relative to the lexical reference route; the reformulation route is deliberately marked shared-backend even though it adds a new identity. Values are task means over the frozen descriptive companion.

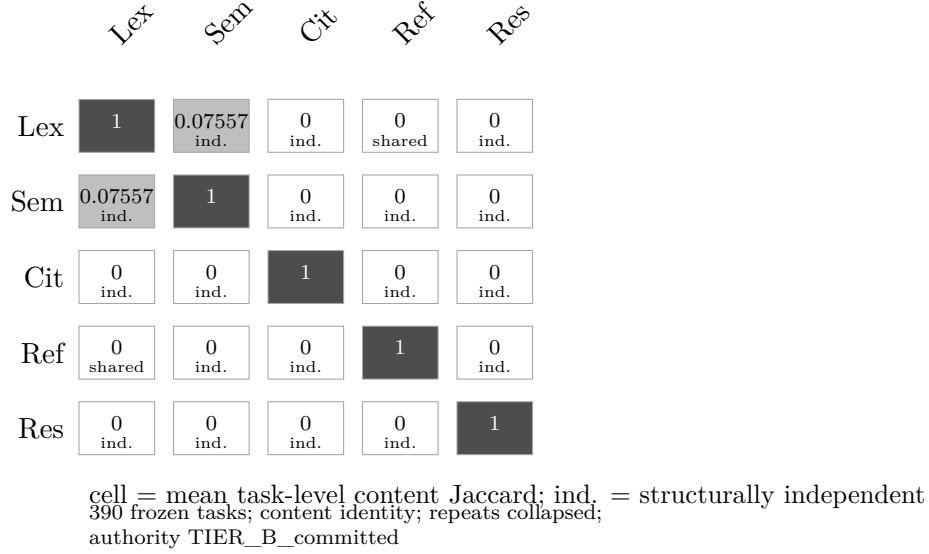


Figure 5: **Figure P2-5.** Mean task-level Jaccard overlap between route captures, computed over content identities rather than source locators, together with the structural independence verdict for each off-diagonal pair. The lexical–semantic overlap is 0.07557; the other off-diagonal mean overlaps are zero in this controlled world. Zero observed overlap is not itself evidence of independence.

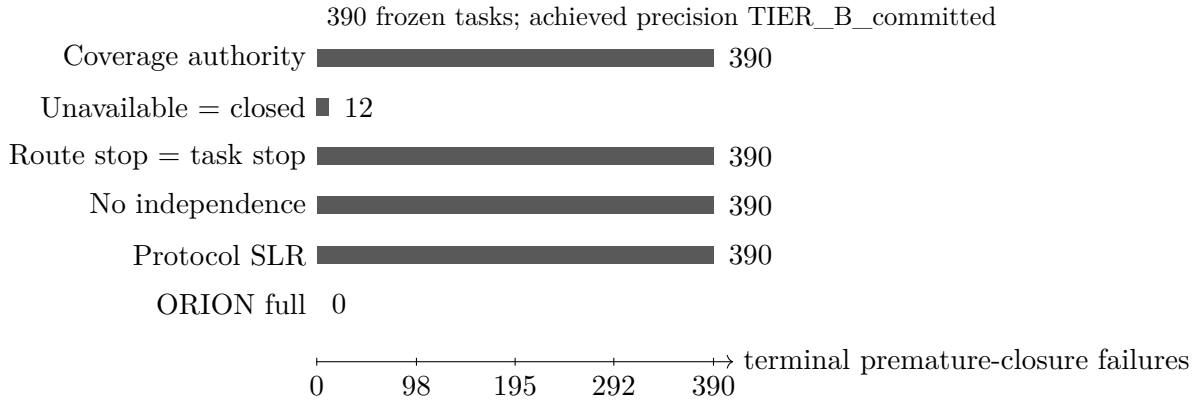


Figure 6: **Figure P2-6.** Terminal premature-closure classifications for selected stopping-safety controls in the frozen 390-task offline companion. The plot is generated from the immutable publication summary. It shows task counts, not an inferential estimate; lower-precedence failure mechanisms can be masked by the evaluator’s terminal-failure precedence.